

IV. EDA-Data Inspection and Analysis

```
[5]: categorical_cols = [  
    'Gender',  
    'Married',  
    'Dependents',  
    'Education',  
    'Self_Employed',  
    'Property_Area'  
]  
  
for col in categorical_cols:  
    unique_vals = df[col].unique()  
    print(f"\nUnique values in '{col}': {unique_vals}")
```

```
Unique values in 'Gender': <StringArray>  
['Male', 'Female']  
Length: 2, dtype: str
```

```
Unique values in 'Married': <StringArray>  
['Yes', 'No']  
Length: 2, dtype: str
```

```
Unique values in 'Dependents': <StringArray>  
['0', '1', '2', '3+']  
Length: 4, dtype: str
```

```
Unique values in 'Education': <StringArray>  
['Graduate', 'Not Graduate']  
Length: 2, dtype: str
```

```
Unique values in 'Self_Employed': <StringArray>  
['No', 'Yes']  
Length: 2, dtype: str
```

```
Unique values in 'Property_Area': <StringArray>  
['Urban', 'Rural', 'Semiurban']  
Length: 3, dtype: str
```

```
length: 3, dtype: str
```

```
[6]: df['Dependents'] = df['Dependents'].replace('3+', 3)

df['Dependents'] = pd.to_numeric(
    df['Dependents'],
    errors='coerce'
).astype('Int64')

print("\nValue counts in 'Dependents' after conversion:")
print(df['Dependents'].value_counts(dropna=False))
```

```
Value counts in 'Dependents' after conversion:
Dependents
0      3
1      3
2      2
3      2
Name: count, dtype: Int64
```

```
[7]: cols = [
    'ApplicantIncome',
    'CoapplicantIncome',
    'LoanAmount'
]

for col in cols:
    print(f"\nStatistics for '{col}':")
    print(f"Mean: {df[col].mean():.2f}")
    print(f"Median: {df[col].median():.2f}")
    print(f"Mode: {df[col].mode().values}")
    print(f"Range: {df[col].max() - df[col].min():.2f}")
    print(f"Variance: {df[col].var():.2f}")
    print(f"Standard Deviation: {df[col].std():.2f}")
```

```
Statistics for 'ApplicantIncome':
Mean: 5550.00
Median: 5250.00
Mode: [3000 3500 4000 4500 5000 5500 6000 7000 8000 9000]
Range: 6000.00
Variance: 3858333.33
Standard Deviation: 1964.26

Statistics for 'CoapplicantIncome':
Mean: 1450.00
Median: 1500.00
Mode: [  0 1500]
Range: 3000.00
Variance: 933888.89
Standard Deviation: 966.38

Statistics for 'LoanAmount':
Mean: 181.00
Median: 170.00
Mode: [100 120 130 150 160 180 200 220 250 300]
Range: 200.00
Variance: 3898.89
Standard Deviation: 62.44
```

```
[8]: high_income = df[df['ApplicantIncome'] > 5000]

print("Applicants with ApplicantIncome greater than 5000:")
print(high_income.head())
```

```
Applicants with ApplicantIncome greater than 5000:
   Loan_ID  Gender  Married  Dependents  Education  Self_Employed \
2  LP003    Male      Yes           2    Not Graduate           Yes
3  LP004    Male      Yes           3      Graduate           No
6  LP007    Male      No            2      Graduate           No
8  LP009    Male      Yes           3    Not Graduate           Yes
9  LP010  Female      No            1      Graduate           No
```

Standard Deviation: 62.44

```
[8]: high_income = df[df['ApplicantIncome'] > 5000]
```

```
print("Applicants with ApplicantIncome greater than 5000:")
print(high_income.head())
```

Applicants with ApplicantIncome greater than 5000:

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	\
2	LP003	Male	Yes	2	Not Graduate	Yes	
3	LP004	Male	Yes	3	Graduate	No	
6	LP007	Male	No	2	Graduate	No	
8	LP009	Male	Yes	3	Not Graduate	Yes	
9	LP010	Female	No	1	Graduate	No	

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	\
2	6000	2000	200	360	
3	8000	2500	250	360	
6	7000	0	220	360	
8	9000	3000	300	360	
9	5500	1500	160	360	

	Credit_History	Property_Area	Loan_Status
2	0.0	Semiurban	N
3	1.0	Urban	Y
6	1.0	Urban	Y
8	1.0	Semiurban	Y
9	1.0	Urban	Y

```
[ ]:
```

RESULT:

Thus, the given program was written and executed successfully.